

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

Forename(s)

Candidate signature

I declare this is my own work.

INTERNATIONAL GCSE

BIOLOGY

Paper 2

Tuesday 29 October 2024

07:00 GMT

Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a ruler with millimetre measurements
- a scientific calculator.

Instructions

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- There are 90 marks available on this paper.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.
- You are reminded of the need for good English and clear presentation in your answers.

Advice

In all calculations, show clearly how you work out your answer.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
TOTAL	



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ANSWER IN THE SPACES PROVIDED**



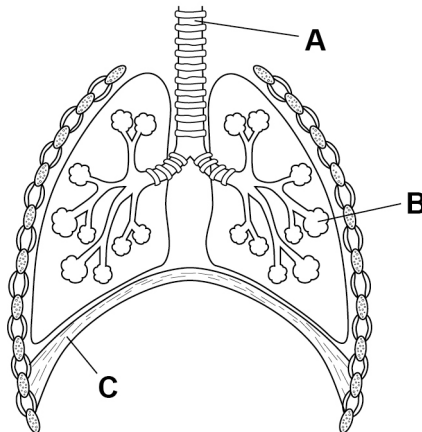
Answer **all** questions in the spaces provided.

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0 1 Breathing allows gases to be exchanged.

0 1 . 1 **Figure 1** shows the breathing system.

Figure 1



Name parts **A**, **B** and **C**.

[3 marks]

Choose answers from the box.

alveolus

diaphragm

oesophagus

rib

trachea

A _____

B _____

C _____

0 1 . 2 How does oxygen move from the lungs into the blood?

[1 mark]

Tick (✓) **one** box.

By active transport

☐

By diffusion

☐

By osmosis

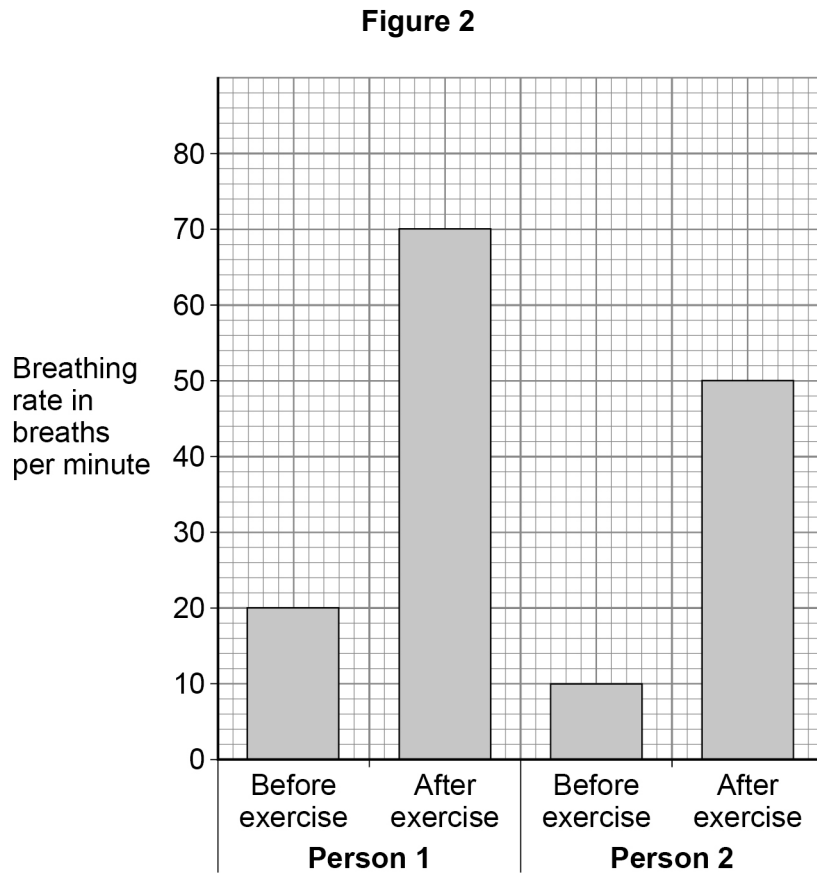
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A scientist investigated the effect of exercise on breathing rate in two people.

Figure 2 shows the results.



0 1 . 3 Calculate the increase in breathing rate for person 2.

Use **Figure 2**.

[2 marks]

Increase in breathing rate = _____ breaths per minute



0 1 . 4

Compare the breathing rate of person 1 with person 2:

- before exercise
- after exercise.

[2 marks]

Before exercise _____

After exercise _____

0 1 . 5

Explain why breathing rate increases during exercise.

[4 marks]

12

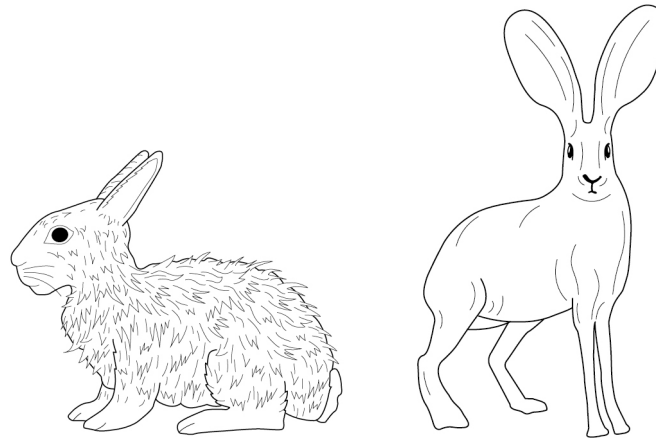
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0 2

Animals are adapted to where they live.

Figure 3 shows two different species of hare.

Figure 3



Hare A

Hare B

0 2 . 1

Hare **A** is adapted for living in a cold climate.

Give **two** adaptations of Hare **A** for living in a cold climate.

Use **Figure 3**.

[2 marks]

1 _____

2 _____



0 2 . 2 All hares have adaptations to help them survive.

Draw **one** line from **each** adaptation to the description of how it helps the hare survive.

[2 marks]

Adaptation	How it helps the hare survive
Eyes on the sides of the head	To detect predators to avoid being eaten
Large feet	To increase surface area to avoid sinking into soft ground
	To provide camouflage to avoid being seen

0 2 . 3 Animals compete with one another.

What do animals compete for?

Tick (✓) **two** boxes.

[2 marks]

Carbon dioxide

☐

Food

☐

Light

☐

Mates

☐

Oxygen

☐

Turn over ►



0 2 . 4 Figure 4 shows a food chain.

Figure 4

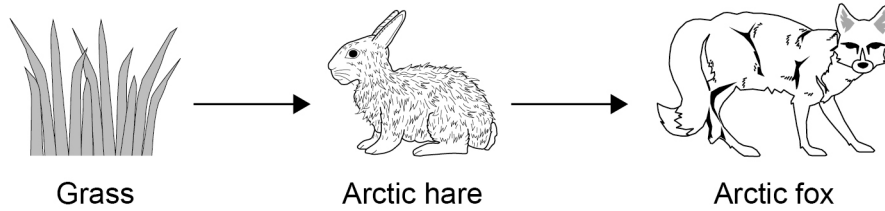
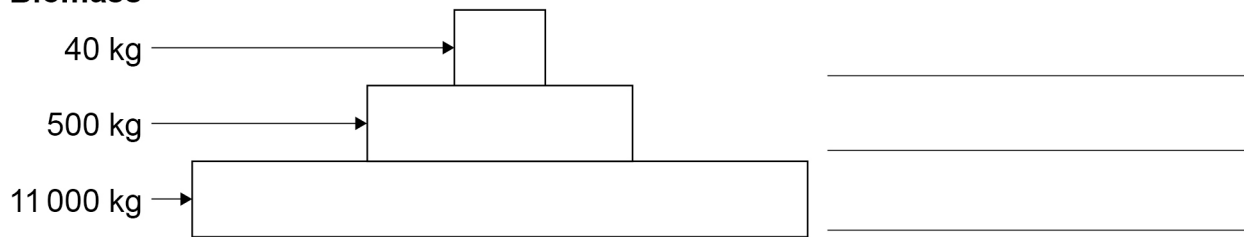


Figure 5 shows a pyramid of biomass for the food chain in **Figure 4**.

Figure 5

Biomass



Complete **Figure 5**.

Write the names of the organisms in the food chain on **Figure 5**.

[1 mark]

0 2 . 5 Some biomass is lost at each stage of the food chain.

Which are **two** ways that biomass is lost?

[2 marks]

Tick (✓) **two** boxes.

As food for the arctic fox

☐

As food for the arctic hare

☐

As a product of photosynthesis in grass

☐

As grass which is not eaten

☐

As waste from the arctic hare

☐

0 2 . 6 Calculate the total biomass lost in the food chain in **Figure 4**.

[2 marks]

Total biomass lost = _____ kg

11

Turn over for the next question

Turn over ►



0 3

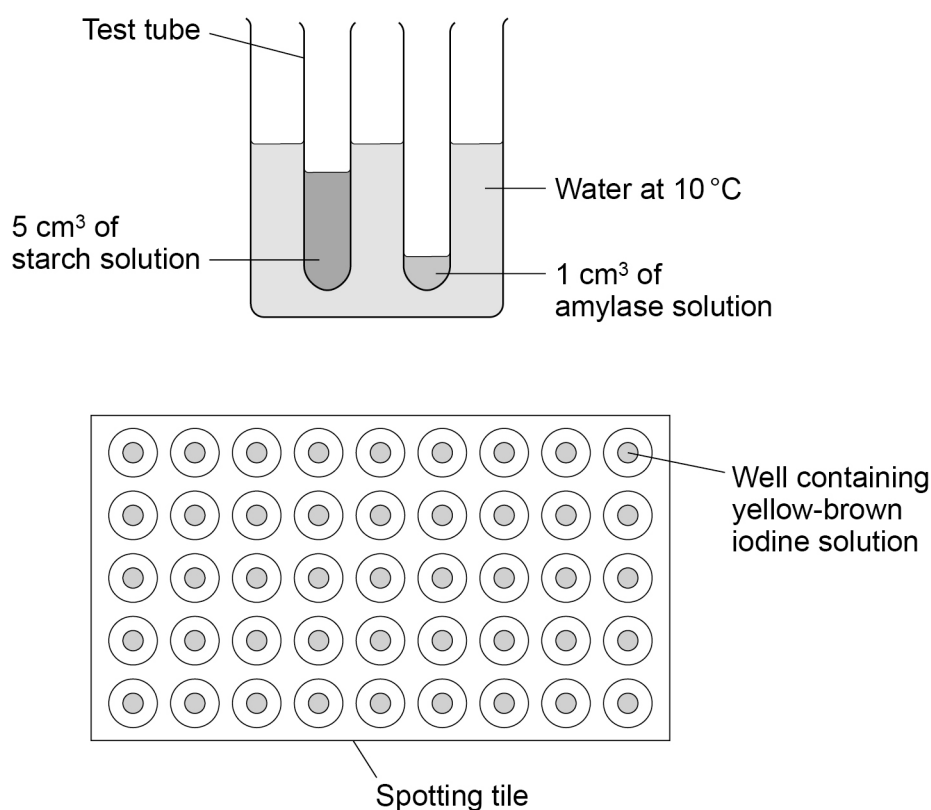
Amylase is an enzyme which breaks down starch.

When starch is added to yellow-brown iodine solution it turns black.

A student investigated the effect of temperature on the activity of the enzyme amylase.

Figure 6 shows the apparatus used.

Figure 6



This is the method used:

1. Set up the apparatus as shown in **Figure 6**.
2. After 5 minutes, mix the starch solution with the amylase solution.
3. Remove 1 drop of the starch-amylase mixture and add to the iodine solution in 1 well of the spotting tile.
4. Record the colour of the iodine solution.
5. Repeat steps 3 and 4 every minute for 8 minutes.
6. Repeat steps 1 to 5 at 15 °C, 20 °C, 25 °C and 30 °C.



03.1

What was the independent variable for this investigation?

[1 mark]

Tick (✓) **one** box.

Size of test tube

☐

Temperature of solution

☐

Time before mixing the solutions

☐

03.2

What are **two** control variables in the student's investigation?

[2 marks]

Tick (✓) **two** boxes.

Size of spotting tile

☐

Time taken for iodine solution to change colour

☐

Volume of amylase solution

☐

Volume of starch solution

☐

Volume of water in beaker

☐

03.3

Which piece of apparatus should the student use to accurately measure 5 cm³ of starch solution?

[1 mark]

Tick (✓) **one** box.

Beaker

☐

Measuring cylinder

☐

Test tube

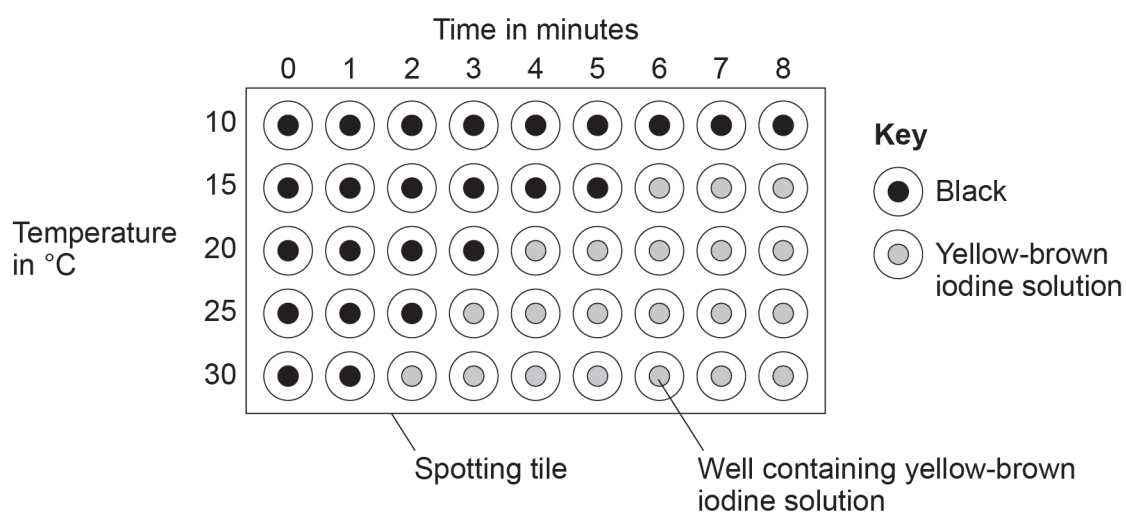
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Figure 7 shows the results of the investigation.

Figure 7



At 20 °C it took 4 minutes for the iodine solution to remain yellow-brown.

0 3 . 4 Why did the iodine solution not turn black at 4 minutes?

[1 mark]

0 3 . 5 How long did it take for the iodine solution to remain yellow-brown at 15 °C?

[1 mark]

Time = _____ minutes



0 3 . 6 The student did the investigation five times and calculated mean values.

Table 1 shows the results.

Table 1

Temperature in °C	Mean time taken for iodine solution to remain yellow-brown in minutes
10	8.0
15	4.0
20	2.8
25	2.2
30	1.8

Complete **Figure 8**.

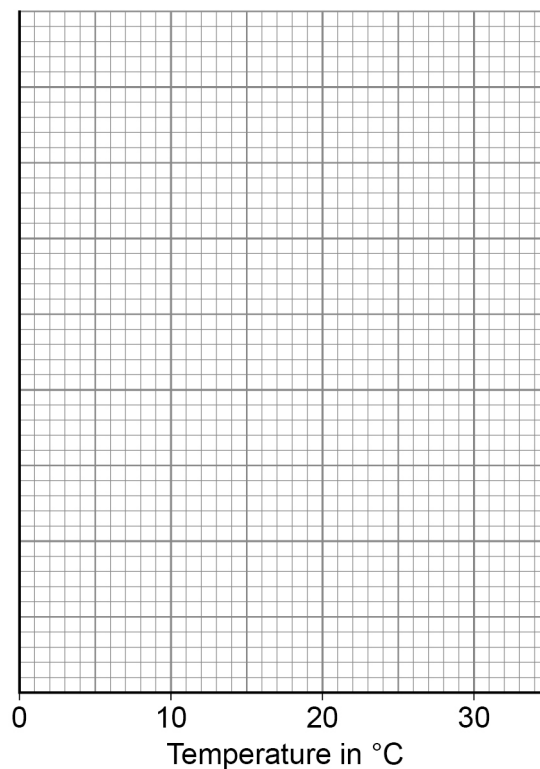
You should:

- use a suitable scale for the y-axis
- plot the data from **Table 1**
- draw a line of best fit.

[4 marks]

Figure 8

Mean time
taken for
iodine solution
to remain
yellow-brown
in minutes



Turn over ►



0 3 . 7 How could the accuracy of the results be improved?

[1 mark]

Tick (✓) **one** box.

Test the solution every 2 minutes

☐

Test the solution every 30 seconds

☐

Use a larger volume of iodine solution

☐

Use a smaller volume of starch solution

☐

0 3 . 8 The investigation was repeated at 60 °C.

After 8 minutes the iodine solution was black.

Give the reason why the iodine solution was black after 8 minutes.

[1 mark]

12



0	4	.	1
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Describe a method to investigate the effect of temperature on the change in mass of potato pieces in a sugar solution.

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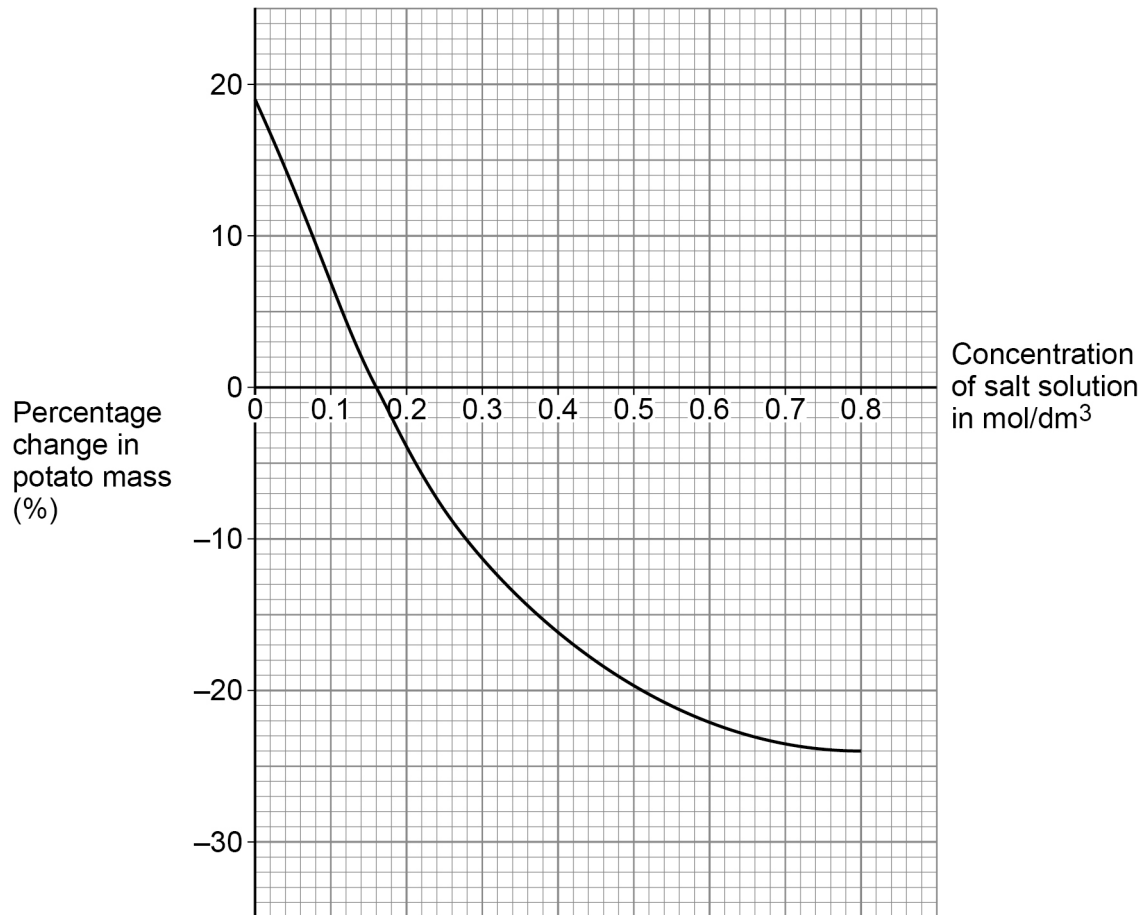
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A different student investigated the effect of concentration of salt solution on the change in mass of potato pieces.

Figure 9 shows the results.

Figure 9



0 4 . 2 Explain why the potato pieces in the 0.1 mol/dm³ salt solution increased in mass.

[3 marks]



0 4 . 3

What concentration of salt solution was equal to the concentration of the solution inside the potato cells?

Use **Figure 9**.

[1 mark]

Concentration = _____ mol/dm³

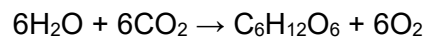
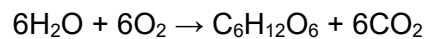
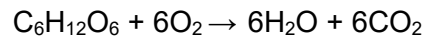
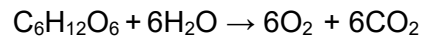
0 4 . 4

The cells in potato respire.

Which equation represents aerobic respiration?

[1 mark]

Tick (✓) **one** box.


☐

☐

☐

☐

0 4 . 5

Chemical **X** can be used to stop aerobic respiration in cells.

The student repeated the investigation into the effect of concentration of salt solution on the change in mass of potato pieces.

The potato pieces were soaked in chemical **X** before they were placed in the salt solution.

Explain what results the student would expect to see.

[2 marks]



0 5

The human heart contains four valves.

0 5

. 1

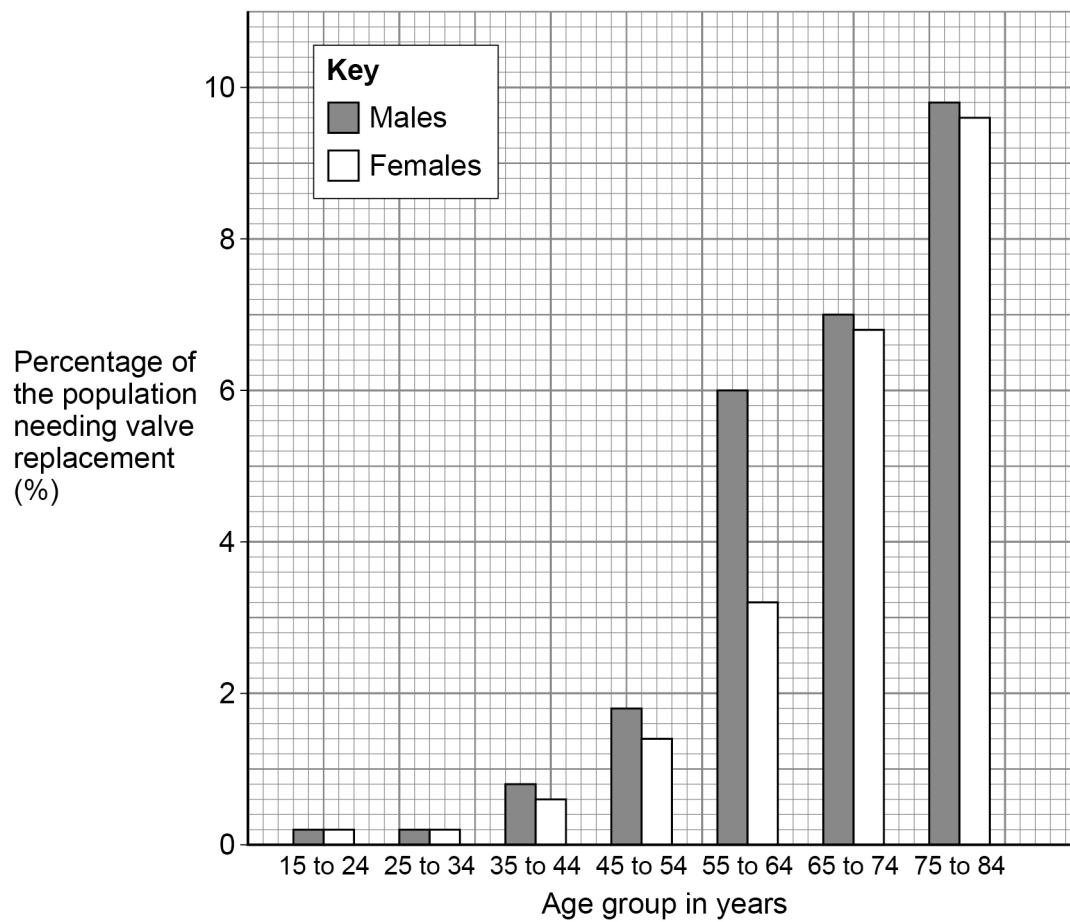
What is the function of valves in the heart?

[1 mark]

Sometimes valves in the heart leak and need to be replaced.

Figure 10 shows data about valve replacement.

Figure 10



0 5 . 2 Give **two** conclusions from **Figure 10**.

[2 marks]

1 _____

2 _____

0 5 . 3 Calculate the difference in the percentage of males compared to females needing heart valve replacements in the age group 55 to 64 years.

Use **Figure 10**.

[2 marks]

Difference (in percentage) = _____ %

0 5 . 4 Suggest **two** symptoms of a leaky heart valve.

[2 marks]

1 _____

2 _____

Question 5 continues on the next page

Turn over ►



A leaky heart valve can be replaced with either a mechanical valve or a biological valve.

Table 2

Mechanical valve	Biological valve
Made of plastic or metal	Made of living tissue
Needs replacing after 10 to 20 years	Needs replacing after 10 to 15 years
The patient needs to take medication to prevent blood clots	The donor valve needs to be a similar tissue type to the patient

Evaluate the use of a biological valve instead of a mechanical valve for the person.

[illegible]

0 6

Large-scale deforestation has damaging effects on the Earth.

0 6 . 1**Table 3** shows deforestation in the Amazon rainforest.**Table 3**

Year	Rate of deforestation in square kilometres per year
2004	27 000
2006	20 100
2008	11 056
2010	8500
2012	4000
2014	6340
2016	8453
2018	9780
2020	10 230

Describe the trend in deforestation in the Amazon rainforest.

Use data from **Table 3**.**[2 marks]**

Question 6 continues on the next page**Turn over ►**

0 6 . 2

Deforestation causes an increase in carbon dioxide in the atmosphere.

An increase of carbon dioxide in the atmosphere contributes to global warming.

Give **two** possible effects of global warming.

[2 marks]

1 _____

2 _____

Deforestation in the Amazon rainforest provides land for maize to be grown.

Maize can be genetically modified.

A gene from mustard plants is inserted to genetically modify the maize.

0 6 . 3

Describe how scientists can genetically modify maize plants.

[5 marks]



0 6 . 4

GM maize is planted to help reduce carbon dioxide levels.

Genetically modified (GM) maize can remove 30% more carbon dioxide from the atmosphere than non-GM maize.

An acre of non-GM maize removes 16 tonnes of carbon dioxide a year.

Deforestation caused an extra 660 million tonnes of carbon dioxide to be released in one year.

Calculate the area of GM maize needed to remove 660 million tonnes of carbon dioxide in one year.

Give your answer in standard form.

[4 marks]

Area (standard form) = _____ acres

0 6 . 5

Some people do **not** agree with the use of GM maize.

Give **one** reason why.

[1 mark]

14

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0 7

Antibiotics kill bacteria.

0 7 . 1

Some antibiotics work by attaching to ribosomes in bacteria.

Explain the effect the antibiotics binding to ribosomes will have on bacteria.

[2 marks]

0 7 . 2Antibiotic **Y** works by attaching to antigens on the surface of bacteria.

The antibiotic has a specific shape that matches the antigens on the surface of bacteria.

Some bacteria have a mutation which makes a protective protein.

The protein attaches to the same antigens on the surface of the bacteria that antibiotic **Y** attaches to.Explain why the bacteria are **not** killed by antibiotic **Y** if they produce the protective protein.**[2 marks]**

Question 7 continues on the next page**Turn over ►**

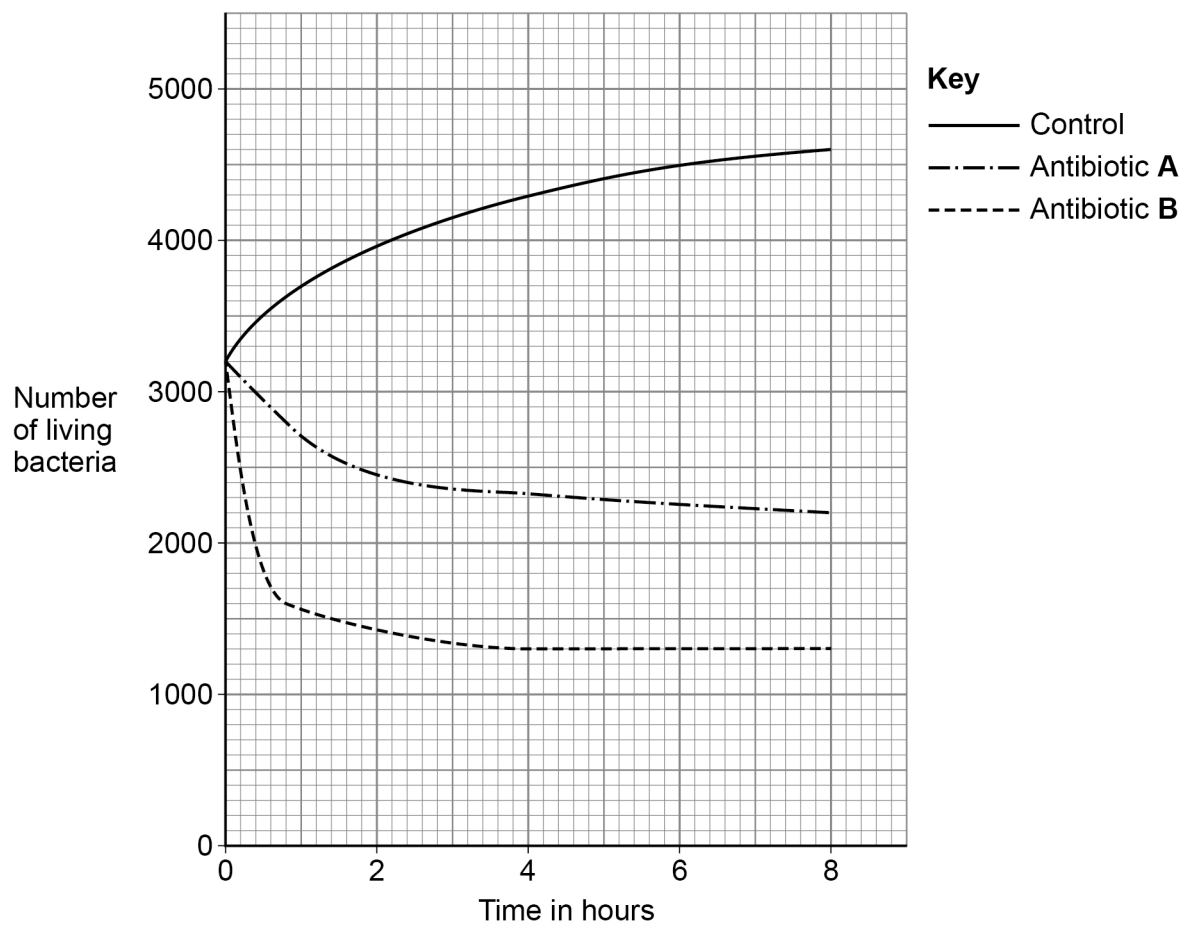
Scientists tested the effectiveness of two new antibiotics on one type of bacteria.

The scientists:

- added each antibiotic to a culture solution containing a known number of living bacteria
- used a control that contained bacteria with no antibiotic added
- counted the number of bacteria that were alive at regular intervals.

Figure 11 shows the results.

Figure 11



0 7 . 3

How many living bacteria were in each culture solution at the start of the investigation?

[1 mark]



0 7 . 4 What is the time of the 50th percentile for the data for antibiotic **B**?

Use **Figure 11**.

[1 mark]

Time = _____ hours

0 7 . 5 Compare the effects of antibiotic **A** with antibiotic **B** on the numbers of living bacteria.

[4 marks]

0 7 . 6 Antibiotic **B** does **not** kill all of the bacteria tested in 8 hours.

Explain why antibiotic **B** can still be effective in treating an infection by the bacteria.

Use data from **Figure 11**.

[2 marks]

Question 7 continues on the next page

Turn over ►



0	7	.	7
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Farmers in some countries add antibiotics to the food they give to animals.

Explain why scientists want the use of antibiotics in animal food to be banned.

[3 marks]

15

END OF QUESTIONS



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